

Temporal Reasoning in Clinical Narratives:

From Information Extraction to Early Disease Detection

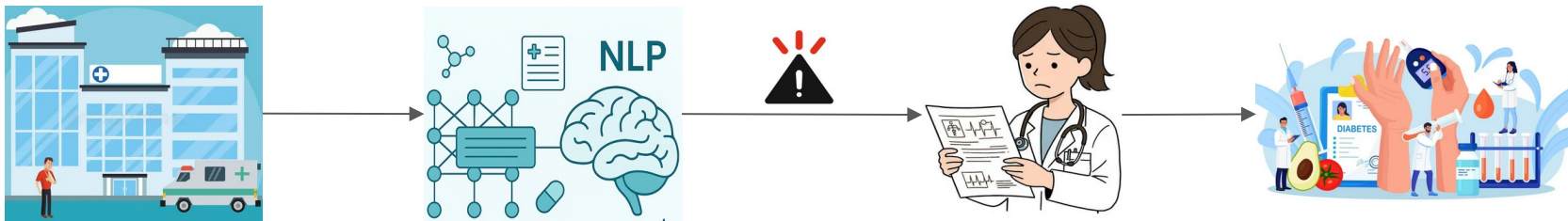


Thanks to Rochana Chaturvedi
(based on her PhD thesis defense)

The Challenge: AI for Chronic Disease Risk Estimation

Type 2 Diabetes (T2D)

- Half Billion Worldwide (and rising)
- Early Diagnosis Important
- Many undiagnosed: 50% UI Health Emergency patients and 23% in U.S
→ Need **Opportunistic Screening** systems (especially for underserved populations lacking routine care)



The Challenge: AI for Chronic Disease Risk Estimation

Current Limitations

- Structured EHR data noisy^[1], lacks crucial context
- Clinical Notes:
 - Manual review not scalable, leads to diagnostic errors
 - Existing AI models do not capture rich temporal context in notes

[1] Hersh et al. "Caveats for the use of operational electronic health record data in comparative effectiveness research." *Medical care* (2013)

Clinical Narratives and Temporally Grounded Methods

Clinical notes capture extensive details including **temporal sequence of events**:

- Symptoms, diagnoses, and treatments *evolve over time*
- Order matters: Drug *after* symptom $\neq$ Symptom *after* Drug

Solution: combine richness of clinical narratives & representational power of advanced AI tools (GNNs, language models) for **structured and contextual reasoning over temporal cues**

Expected Impact: Advancing the state of the art in clinical NLP and healthcare AI

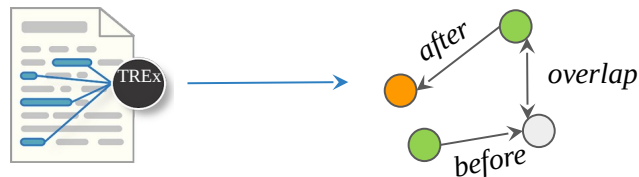
- (1) End-to-end entity-temporal-relation extraction
- (2) Real healthcare decision support applications, e.g. opportunistic screening of T2D

Presentation Roadmap

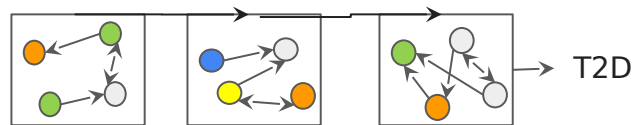
**Visit-Level Temporal cues for
T2D-risk prediction**



Temporal Relation Extraction



**Risk Prediction with
Hierarchical Temporal
Graphs**



Conclusion and Future Directions

Part 1

Leveraging Visit-Level Temporal Cues For Type 2 Diabetes Prediction





- Longitudinal EHR >10 years (Jan 1, 2010, to Jul 31, 2021)
- Notes split into EPIC and CERNER (until Fall 2020) subset
- Inclusion Criteria:
 - Adults
 - At least 2 outpatient visits in 24 months
- Exclusion Criteria:
 - Diagnosis of Type 1 diabetes or other types of diabetes

Data Filtering: UI Health

* First Filtration Criteria: Only consider patients with some notes

Notes | T2D Patients | NoD Patients

▶ Initial Data

3708168 | 15140 | 164910

▶ Drop Pre-Diagnosed (Earliest_Dx_Date)

3146140 | 4138 | 164910

▶ Demographic Filters (RACE not in [H, NA, AS]; ETHNICITY/GENDER != NI)

2909417 | 3878 | 149673

▶ Note Type and Author types Filter

1451626 | 2369 | 108635

▶ note length >100

1050839 | 2302 | 106222

▶ LLM Filter (adjusts Earliest_dx_Date)

1039043 | 1717 | 106222

Data Cleaning Prompts

Prompt

You are a helpful assistant who can read and identify required information from given clinical notes either mentioned explicitly or in ICD code format.

Required information: a diagnosis of **Diabetes** for the given patient.

Please reply **True** if there is an explicit diagnosis mentioned in the note.

Please answer **False** if:

- there is no diagnosis,
- the note negates the diagnosis (e.g., “no diagnosis of diabetes” or “diabetes: negative”),
- if the diagnosis is associated with someone other than the patient (e.g., family).

Do not output anything other than **True/False**.

NOTE: [NOTE_TEXT]

Output:

Example prompt instructing an LLM to identify **explicit Diabetes diagnosis** in a clinical note

- Paraphrased version of above prompt with additional instruction to ‘*not infer a diagnosis if not explicitly mentioned*’
- If answer is yes, move earliest diagnosis date back to current note date
- Intersection of output of two prompts considered (yes if both answer yes, else no) based on manual checking of 50 examples (notes)

Event Extraction from Unstructured Clinical Notes

with cTAKES^[1]

- extracts clinical concepts from text
- maps synonymous concepts to unique identifiers (CUI) in UMLS^[2] clinical knowledge graph

Patient with a history of diabetic foot and foot checkup. Patient states the toenails are thick, elongated, unable to trim on her own. Patient denies any recent history of trauma. Debrided nails today.

ANATOMY	DISEASE/DISORDER	SIGN/SYMPTOM	MEDICATION

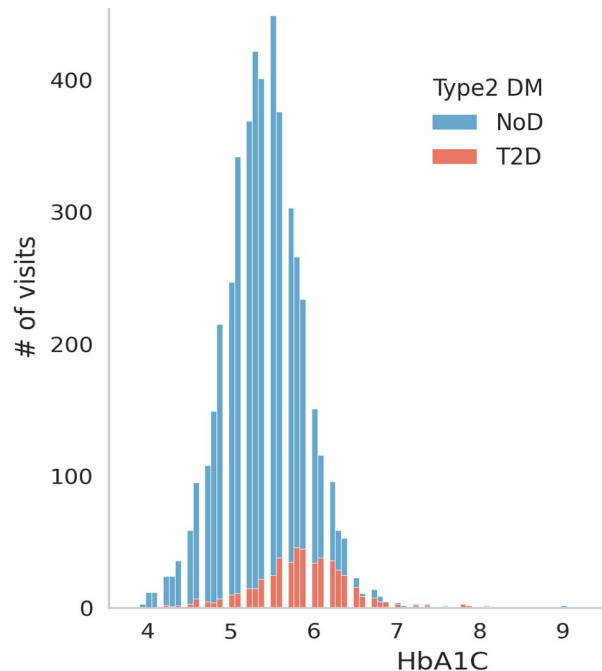
[1] Savova et al. "Mayo clinical text analysis and knowledge extraction system (ctakes): architecture, component evaluation and applications," JAMIA. 2010.

[2] Bodenreider "Using UMLS semantics for classification purposes." *AMIA Symposium*. 2000.

Leveraging Structured Data

HbA1C Time-series

- Test for T2D
- Diabetic if $\text{HbA1C} > 6.5^{[1]}$
- Noisy and sparse (missing from 56.8% observations) leading to missed diagnoses



Demographic Attributes

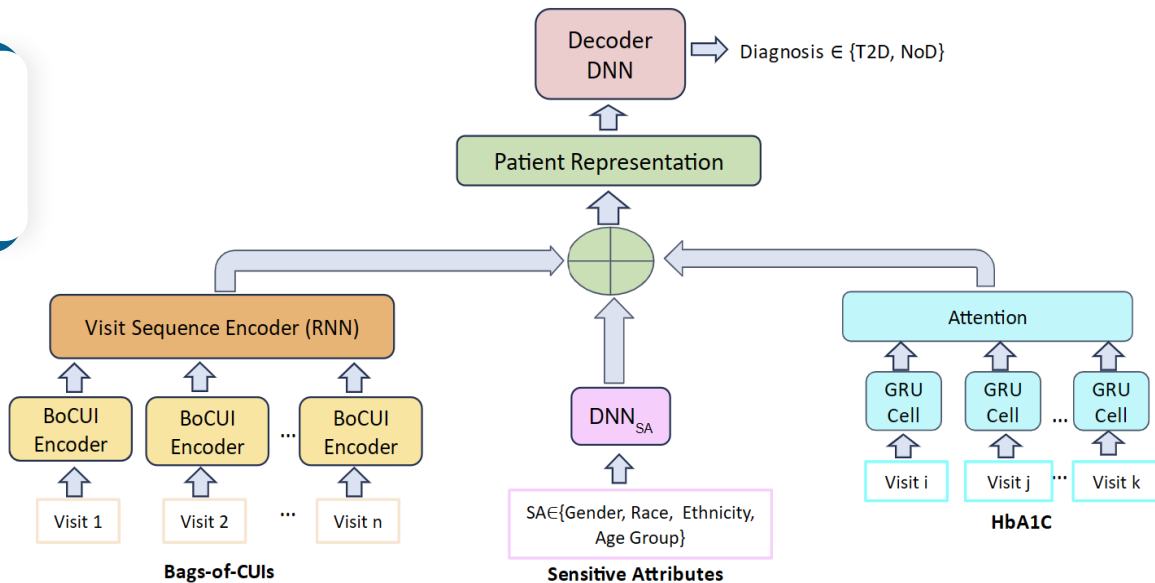
[Details »](#)

Method and Result Summary

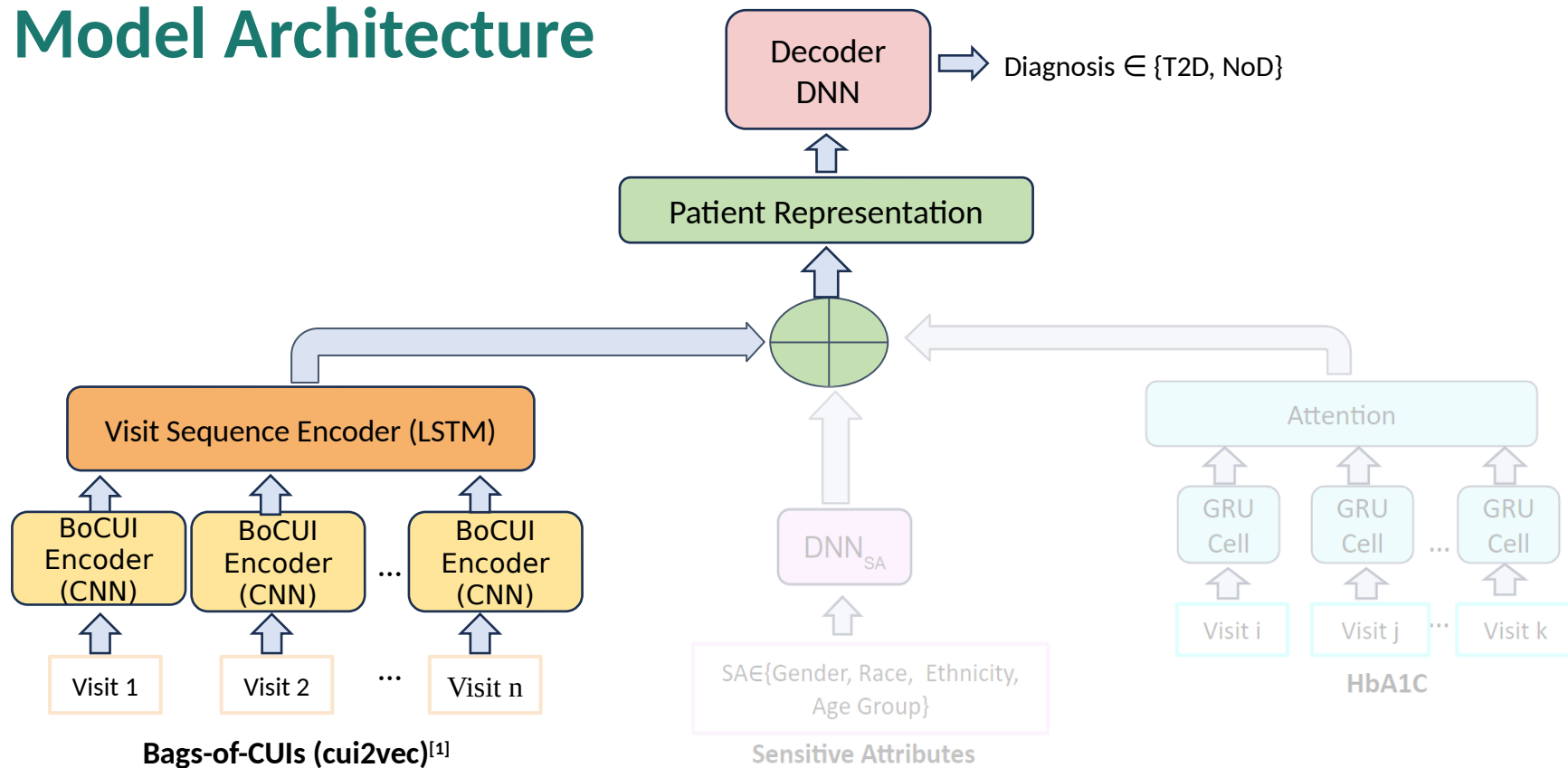
Clinical concepts from notes improve performance over structured EHR

Additional gains from **Temporal modeling** of events **over visits**

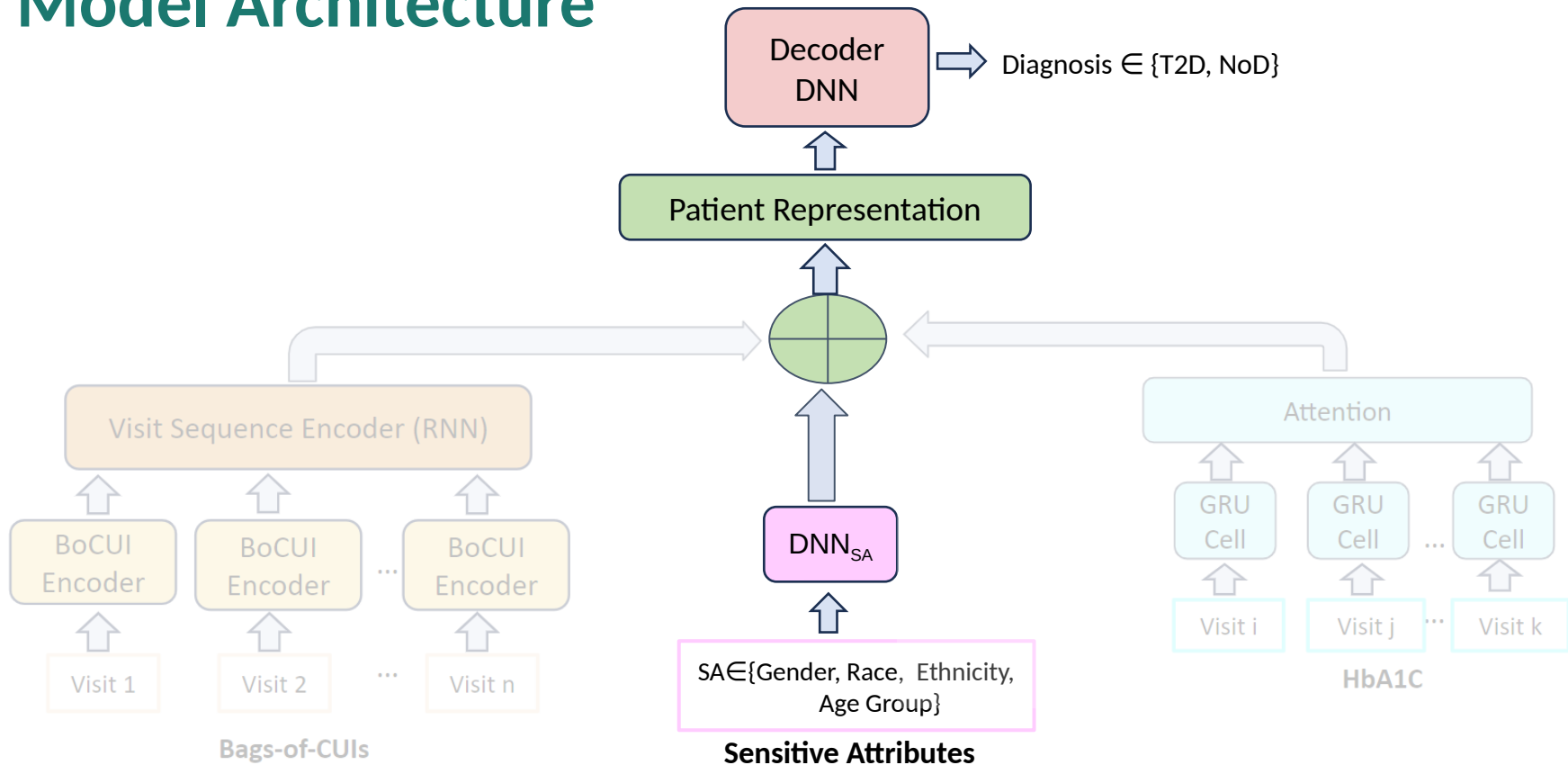
Evaluation Details



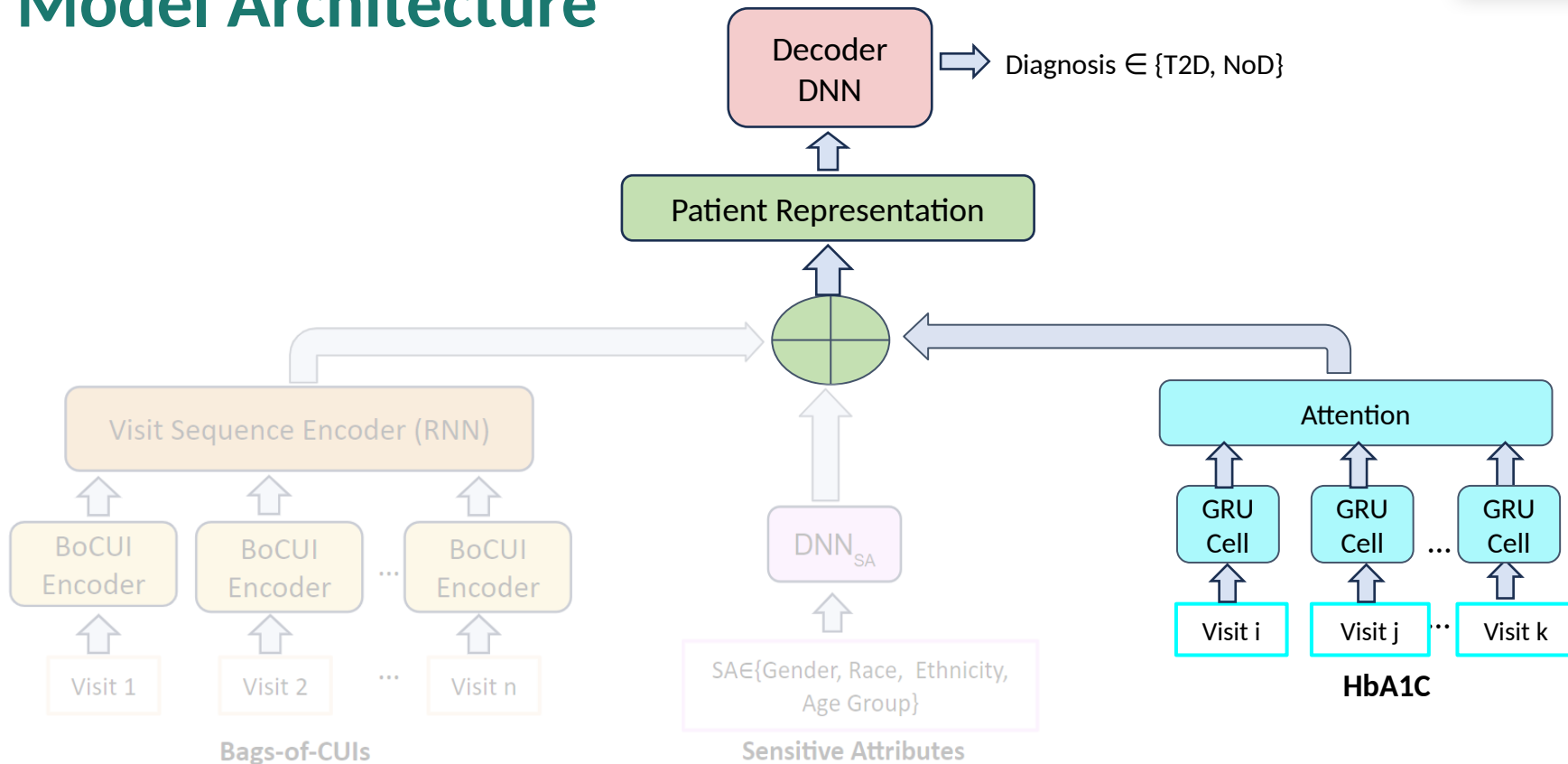
Model Architecture



Model Architecture



Model Architecture



Evaluation Summary

Modality	Model	Macro F1	Type 2 Diabetes		No Diabetes	
			Precision	Recall	Precision	Recall
Structured	HbA1C _{ts} (RNN+Attention)	65.3	35.2	37.4	94.7	94.2
Unstructured Bag-of-CUIs	LR	68.2	50.9	33.7	94.5	97.2
Unstructured Time Series of CUIs	CNN-LSTM	74.2	46.0	62.7	96.7	93.8
Hybrid	CNN-LSTM + HbA1C _{ts}	69.7	59.6	33.7	94.6	98.1
Number of Patients		1063	83		980	

- HbA1C time series model has best performance among structured models

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- Clinical concepts improve performance
- Temporal modeling of concepts gives best performance

Evaluation Summary

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- Adding structured to unstructured reduces recall of T2D class

Subgroup Biases

- Lower performance for Females despite being majority
- Hybrid models with HbA1C test benefit the White group only

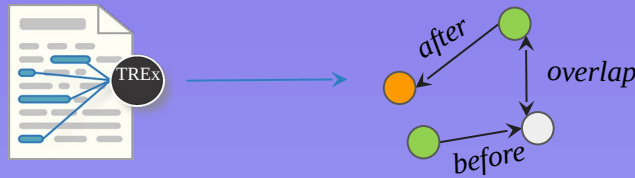
Key Research Questions

RQ2: Can fine-grained temporal information improve patient representation hence risk prediction?

RQ1: Can we improve temporal relation extraction from clinical notes?

Part 2

End-to-end Temporal Relation Extraction from Clinical Text (ACL 25)



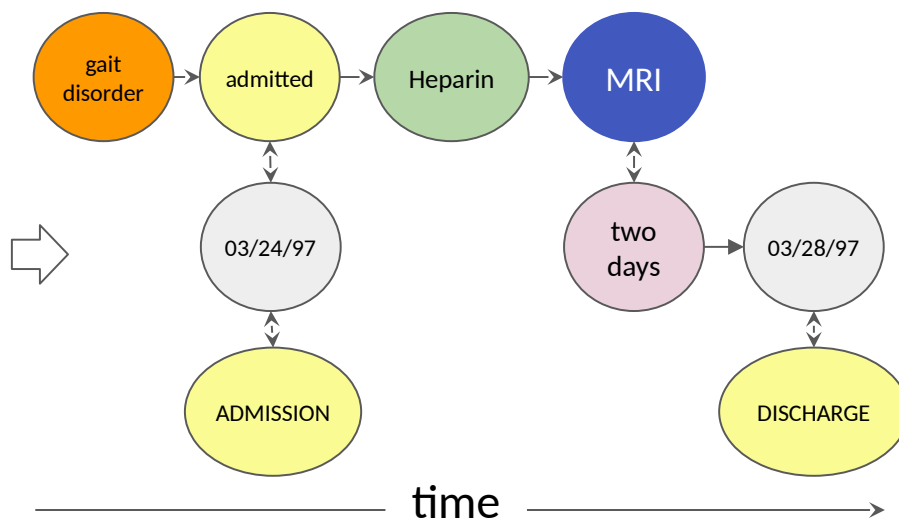
Temporal Relation Extraction (TReX)

ADMISSION DATE: 03/24/97

DISCHARGE DATE: 03/28/97

The patient was admitted for her gait disorder ...

She was initially placed on heparin for two days, until her MRI results were obtained.



TLINK

- ↔ Overlap
- Before
- ← After

Events

PROBLEM

OCCURRENCE

TREATMENT

TEST

TimEx

DATE

DURATION

Challenges



- Clinical jargon
 - Abbreviations: b.i.d (twice a day)
 - Synonyms: T2DM/T2D/DM2/Diabetes Type 2
 - Homonyms (context shapes meaning):

DM2 ~ Type 2 Diabetes/Dystrophia Myotonica

Type 2

- Implicit temporal dependencies:

(A before B) & (B before C) $\rightarrow$ (A before C)

- Long documents, $O(n^2)$ pairs challenge for transformer-based models

Solution: Graph-based Temporal Relation Extraction (GRAPHTREX)

- Large clinically pre-trained language models (LPLM)
 - Learn local, contextualized representations from text but
 - Don't capture global, long-range dependencies
- Graph Neural Networks (GNN)
 - Excel in relational reasoning, but
 - Need semantic-rich embeddings from LPLMs

Our Solution: GRAPHTREX

- Fuses LPLMs with heterogeneous GNNs for contextual and structured extraction and
- Applies sliding windows to handle long document context

Heterogeneous Document-Level Graph

Innovative Local and Global Landmarks

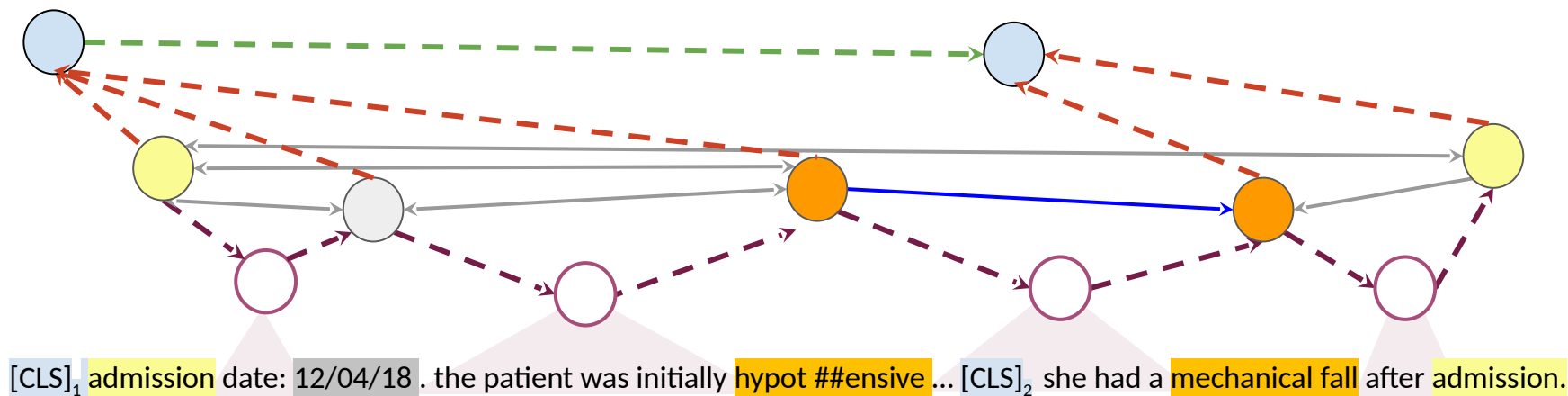
○ Context Nodes: capture context between two entities

● Window Nodes: bridge distant entities

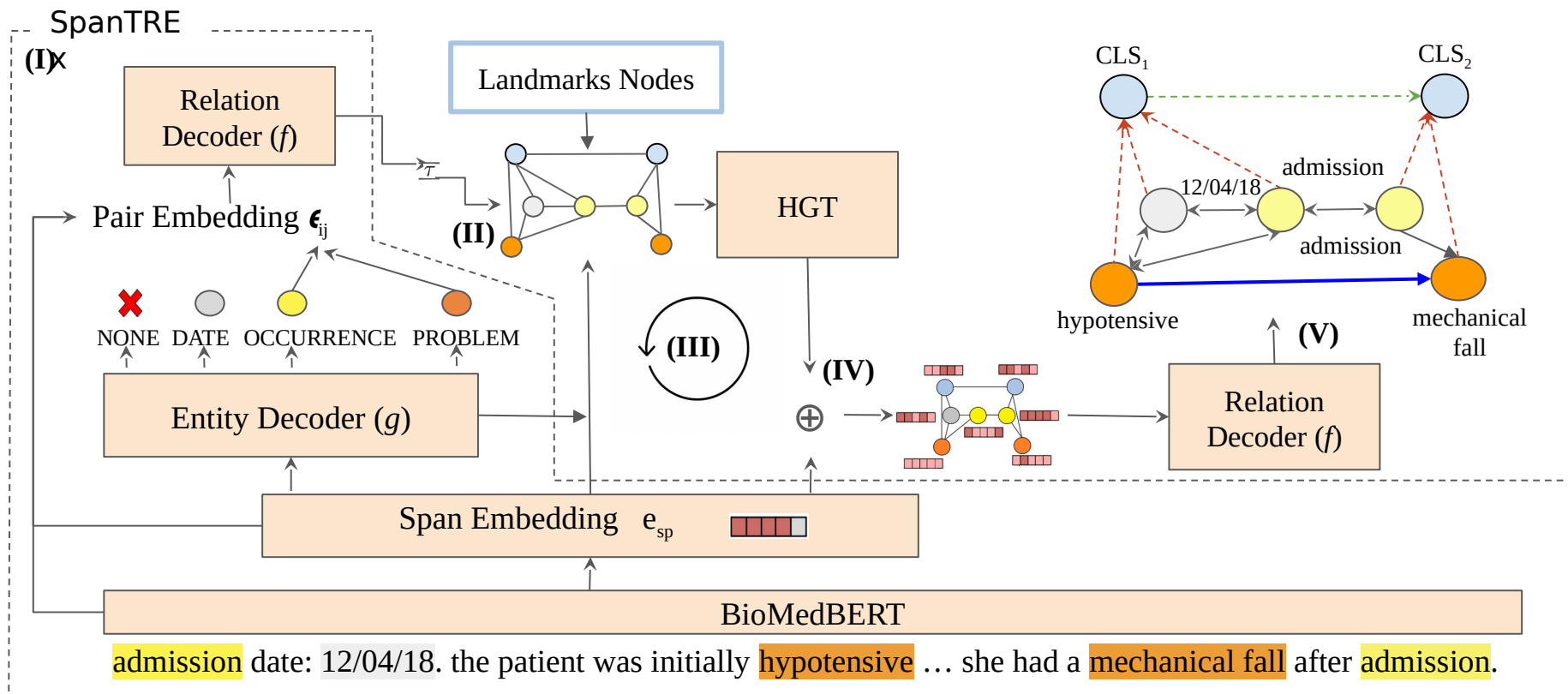
Temporal Edges

→ before HGT

→ after HGT



Graph-based Temporal Relation Extraction (GraphTREx)



Dataset Overview



I2B2 2012 Corpus^[1]

- 310 discharge summaries
- ~35K entities
- ~61K relations



E3C Corpus^[2]

- 84 clinical case reports
- ~5K entities
- ~4K relations

[1] Sun, Rumshisky & Uzuner "Annotating temporal information in clinical narratives". JBI. 2013.

[2] Magnini et al. European clinical case corpus. In European Language Grid: A Language Technology Platform for Multilingual Europe. 2022

Tempeval ... evaluation: TLink F1

Temporal order can be described in different ways:

$$A < B \leftrightarrow B > A$$
$$A < B, B = C \leftrightarrow A < C, B = C$$

$$P = \frac{G_{sys}^- \cap G_{Gold}^+}{G_{sys}^-}$$

$$R = \frac{G_{Gold}^- \cap G_{sys}^+}{G_{Gold}^-}$$

$$F_1 = 2 * \frac{P * R}{P + R}$$

G^+ : closure of G that makes implicit relations explicit (transitivity, invertibility)

G^- : reduced G after removing redundant relations

Evaluation

Baselines	Model	I2B2 Tempeval ^[3] F_1	E3C Micro F_1
	Rules +ML Hybrid ^[1]	63.4	NA
	SPERT ^[2]	61.1	13.6
Ours	SPAN TREX	66.6	22.6
	GRAPH TREX	68.8	23.5

- **New state-of-the-art** on I2B2 2012 end-to-end relation extraction > 5% improvement
- Consistent improvements on E3C corpus

[1] Xu Y, Wang Y, Liu T, Tsujii J, Chang EI. An end-to-end system to identify temporal relation in discharge summaries: 2012 i2b2 challenge. JAMIA. 2013.

[2] Eberts M, Ulges A. Span-based joint entity and relation extraction with transformer pre-training. ECAI. 2020.

[3] UzZaman N, Allen J. Temporal evaluation. ACL. 2011.

Performance Across Relation Types

Corpus	RelType	#TLinks	F1
I2B2 2012	<i>Overlap</i>	7155	59.27
	<i>Before</i>	13453	75.05
	<i>After</i>	13453	74.94
E3C	<i>Overlap</i>	1560	24.23
	<i>Before</i>	793	21.88

What makes GraphTREx Work: Landmark Nodes

Entity	Landmark Nodes		% IMP	
	Context	Window	I2B2 2012	E3C
✓			-3.3	0.8
✓	✓		1.9	4.1
✓		✓	1.3	3.7
✓	✓	✓	2.3	1.9

- I2B2 2012: Both landmark nodes enhance performance
- E3C: short documents, window nodes create noise

Additional robustness experiments across base encoders, pair embedding approaches

[Details »](#)

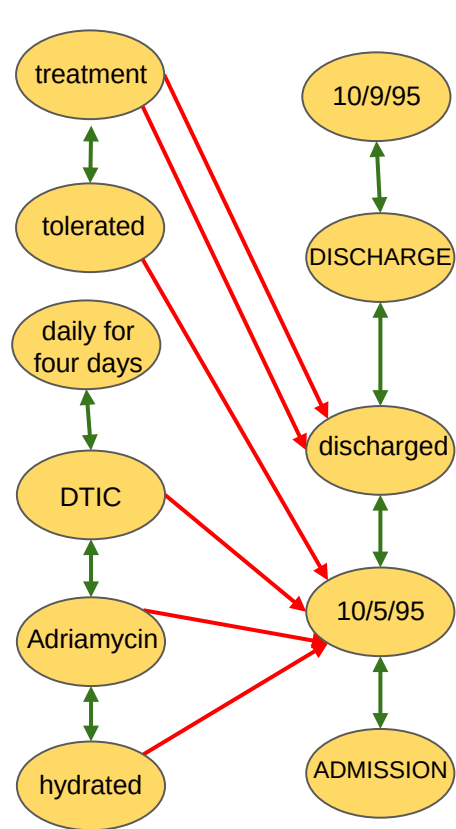
Performance Across Distant Entities

Pair Distance	% IMP
d=0	4.5
d>0	6.7
d>1	8.9

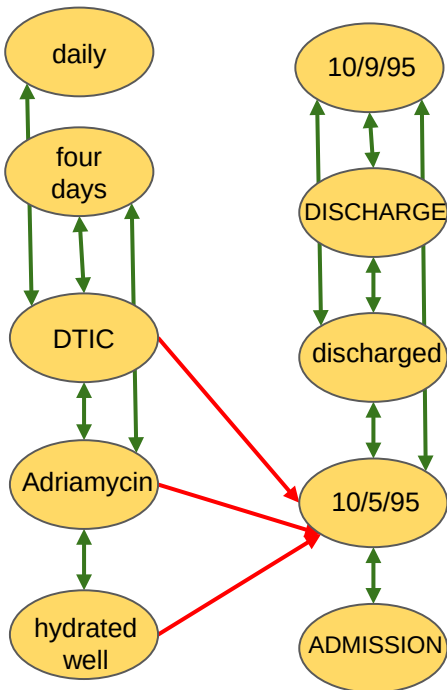
Consistent improvement (%IMP) from GRAPHTREX over SPANTREX across:

- d=0: Nearby pairs (within the same window)
- d>0: Distant pairs (spanning different windows)
- d>1: Farther pairs (across two or more windows)

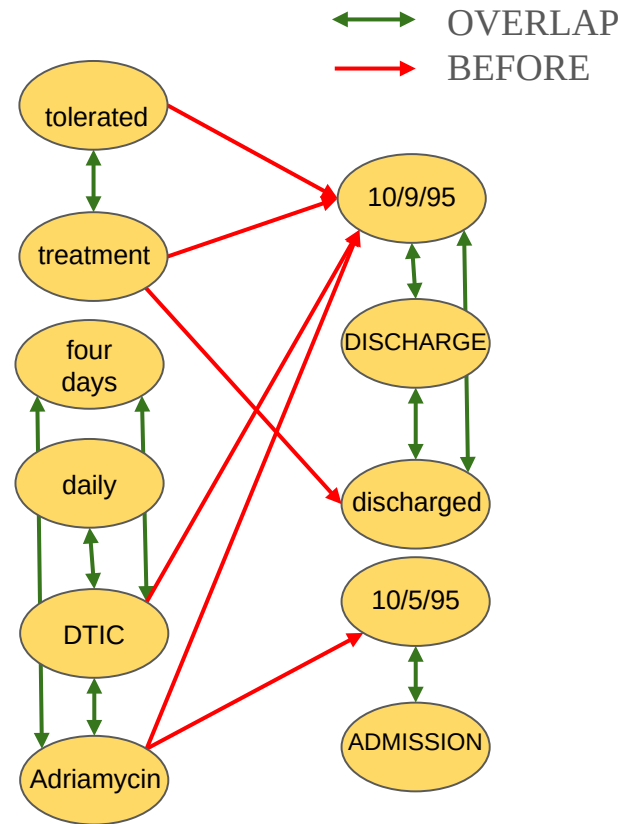
Case Study: Temporal Graphs from a Snippet from I2B2 2012



GROUND-TRUTH



SPANTREX



GRAPHTREX

Key Contributions

Introduce GRAPH TREX, a novel integration of GNNs and LPLMs with innovative landmark nodes

- SOTA on End-to-end clinical temporal relation extraction
- Especially improves predictions on long-range relations

Groundwork for Event Sequencing in Clinical Narratives

- Can TREx models be integrated into clinical decision support to improve patient representation hence health-risk prediction? (RQ2)
 - How well do TREx models transfer to a new setting with real-world data

Transferring TReX Models: from I2B2 to UI Health

Key Differences in Data Distribution:

I2B2: subsections of discharge summaries

UI Health: Progress notes, family medicine, endocrinology notes, etc.

Institutional Factors: Different hospitals, EHR, authors

Patient Demographics



UI Health Sample

- 9 clinical notes
- 2 patients
- Expert Collaboration

Annotation Categories

Deidentification

PHI removal
(HIPAA)
Realistic
substitution

Entity Annotation

Events, TimEx

*Silver Standard with
SpanTREx*

Annotation
guidelines extended
from I2B2

Coreference

*Silver standard
linking: Events
(Metamap), Date
(Microsoft
Recognizers text)*

Timeline (TLINK)

Temporal relations
Cross-document

Entity/Coreference Annotation Interface

[» Back](#)

NarrativeTime
annotation tool

1. Select the annotation task to perform:

Events & coreference
Timeline annotation

2. Load the text for annotation

Import file(s)
< 3046 >

3. Annotate!

Clear all Save Export
Toggle quick help

Timex

Event

tab , PO , BID levETIRAcetam 250 mg oral tablet 250 mg = 1 tab , PO , BID Problem list : Active Problems (10) Constipation Elevated liver function tests Epilepsy Glucose intolerance
Hearing loss Ingrown toenail Ingrown toenail Obesity Physiologic obstructive defecation Rectal bleeding Histories Past Medical History : Resolved Bradycardia (6217HC7Z - RD08 - 0132 -
NPP3 - 6B2M0647J68V) : Onset in 2016 at 20 years . Resolved . Hard of hearing (WP234G0S - 80X5 - 009P - 8QF0 - 81F191V3619H) : Onset in 2009 at 13 years . Resolved . Seizure (990916998) : Onset in 2007 at 11 years . Resolved . Issue of medical certificate (543534624) : Resolved . Family History : CA - Lung cancer Grandmother (hx) Grandmother (hx) Asthma
Grandmother (hx) Grandmother (hx) Procedure history : colonoscopy on 12 / 25 / 2017 at 22 Years . RIGHT BONE ANCHORED HEARING AID WITH BAHAAATTRACT on 6 / 01 / 2017 at 21
Years . Tonsillectomy (SNOMED 2nd ed P - 1100) in 2003 at 8 Years . , colonoscopy : 12 / 25 / 17 RIGHT BONE ANCHORED HEARING AID WITH BAHAAATTRACT : 06 / 01 / 17
Tonsillectomy : 2003 Social History Alcohol Details : DENIES , No Details : DENIES , No Employment / School Details : Employed Details : Part time , Activity level : Heavy physical work .
Highest education level : Some college . Operates hazardous equipment : No . Home / Environment Details : Lives with Mother . Living situation : Home / Independent . Nutrition / Health Details : Regular Sexual
Details : Sexually active : No . Substance Abuse Details : DENIES , Are you ready to receive education material on drug cessation ? No . Details : DENIES , Are you ready to receive education material on

	ID	CoRef	Type
11 years	6262	6260	DATE
Lung cancer Grandmother	6282		PROBLEM
Asthma Grandmother	6292		PROBLEM
colonoscopy	6304		TREATMENT
12 / 25 / 2017	6306		DATE
22 Years	6312	6306	DATE
RIGHT BONE ANCHORED HEARING AID WITH BAHAAATTRACT	6315		TREATMENT
6 / 01 / 2017	6324		DATE

Infinity

-Infinity

Timeline Annotation Interface

» Back



NarrativeTime
annotation tool

1. Select the annotation task to perform:

Events & coreference
Timeline annotation

2. Load the text for annotation

Import file(s)
< 11051 >

3. Annotate!

Clear all Save Export
Toggle quick help

[B]

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Note Date : [2016 - 04 - 16] [B] Service : [Emergency] [B] Patient : GARCIA, LUCIANO MRN : 75107991 FIN : 18384251 - 3542 Age : 41 years Sex : MALE DOB : [11 / 05 / 1974] [B] Associated Diagnoses : [Acute otitis externa of right ear] [B] ; [Acute otitis media , right] [B] ; [Hypertension] [B] Author : Rajesh APN , Madeline Basic Information Time seen : Date & time [4 / 15 / 2016] [B] [9 : 12 : 00 PM] [B] . History source : Patient . Arrival mode : [Walking] [B] . History limitation : None . History of Present Illness The patient [presents] [B] with [right ear pain] [B] x [1 week] [B] . PMH [HTN] [U] . Pt [states] [B] he thought he had a [cold] [U] , taking [nyquil] [B] without relief . [Denies] [ha / dizziness] [B] , [denies] [f / c / n / v / d] [B] , [denies] [cp / sob] [B] , [denies] [change in bowel / bladder pattern] [B] , [recently traveled] [U] back from Texas , ear pain began after coming home , [not taking any meds for pain] [B] , did not take [Losartan] [B] today . [Pain aching , throbbing] [B] , [denies] [ringing] [B] , [denies] [drainage] [B] . Review of Systems Additional review of systems information : All other systems reviewed and otherwise negative . Health Status Allergies : Allergic Reactions (All) Severity Not Documented [Lisinopril] [U] : No reactions were documented . Canceled / Inactive Reactions (All) [NKA] [U] . Medications : (Selected) Prescriptions Prescribed [Right thumb spica splint] [B] with velcro : Right thumb spica splint with velcro , See Instructions , [right thumb metacarpal fracture] [B] V54 . 19 wear [regularly] [B] for thumb support for [six weeks] [B] , Dispense Supply : 1 EA , 0 Refill (s) , Effective Date [06 / 19 / 14] [12 : 51 : 00] [B] losartan 50 mg oral tablet : 50 mg = 1 tab , PO , [DAILY] [B] , Dispense Supply : 30 tab , Refills : 5 , Effective Date [06 / 11 / 14] [18 : 48 : 42] [B] , [Family Medicine] [B] 9180 W 44TH COURT TWIN OAKS , IL 60115 . Past Medical / Family / Social History Medical history : Resolved Hypertension (84980841) : Resolved .. Surgical history : [gun shot] in [1987] [B] at 13 Years .. Family history : [Diabetes mellitus type 1 Mother] [U] (hx) . Problem list : Active Problems (4) [Diverticulitis] [U] Family history -

[11 / 05 / 1974]

[Acute otitis externa of right ear] [B] ; [Acute otitis media , right] [B] ; [Hypertension]

[4 / 15 / 2016]

[9 : 12 : 00 PM]

[Walking]

[presents]

[right ear pain]

[1 week]

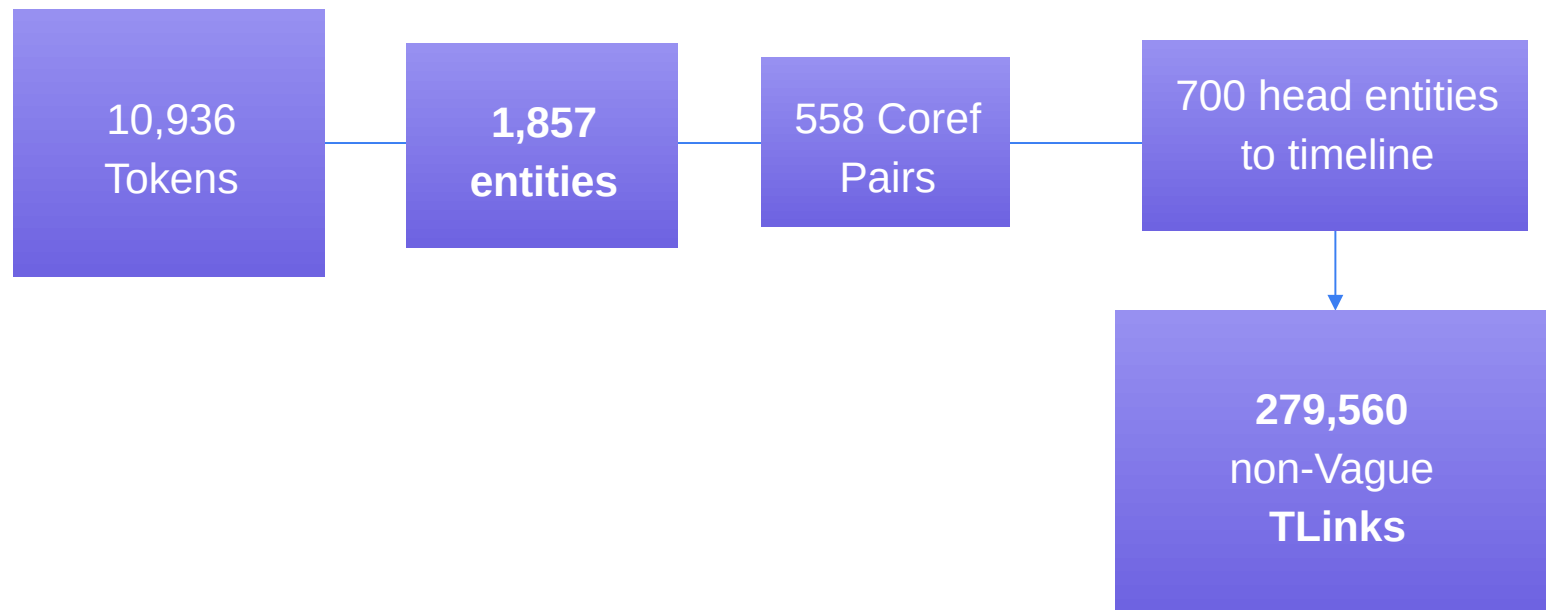
[HTN]

100

[B] -7:0 [right ear pain]

310

Annotation Statistics



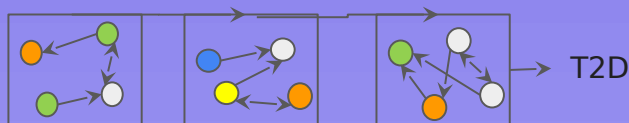
Evaluation

Model	Event (F1)	Timex (F1)	TLink (F1)
SPANTREX	74.4	69.5	38.0
GRAPHTREX	73.5	68.4	52.0

- **Entity Extraction:** SPANTREX slightly outperforms
- **Temporal Relations:** GRAPHTREX achieves **14% improvement** in TLink F_1

Part 3

Disease Risk Prediction with Temporal Graphs



Complementary Perspectives

RQ2 : Can fine-grained temporal information improve patient representations hence risk prediction?
Hit-GNN (similar to GraphTrex)

RQ3 : How to adapt LLMs demonstrating excellent reasoning abilities in low-resource settings?

ReVeAL: Reasoning with Verifier Aided Labeling

Inference time scaling framework for Explainable Disease Prediction with LLMs



Large Reasoner (Llama3.1 8B)



- Generates predictions **with explanations**
- Scaling with **diverse reasoning paths**
 - 5/training example
 - 10/test example

Small Verifier (Llama3.2 1B)



- Validates reasoning from larger model
- **Fine-tuned** with LoRA
 - Input + reasoning from large model
 - Assigns confidence scores (normalized token probability of correct/incorrect)
 - Training with hard cases (reasoner answers mix of true/false over different paths)
- Inference with top-k (3) majority vote

LLM Baselines

Comparisons with Llama3.1-8b: zero-shot/self consistency^[1] (10 reasoning paths)

Comparisons with Llama3.2-1b: zero-shot/fine-tuned

Prompting Strategy:

- Prompts include
 - Patient demographic attributes (age, race/ethnicity, gender),
 - Last n notes with visit dates (chronological order)
- **Reasoner:** asked to generate explanation + risk (True/False)
- **Fine-tuned model:** only generates risk
- **Verifier** (in ReVeAL): reads above inputs + reasoner output (explanation & classification) and generates verification (correct/incorrect)

Prompts

System Prompt: Fine-tuned Verifier

You are a medical reasoning assistant. Given a sequence of clinical notes from a patient and an analysis predicting the risk of an imminent Type 2 Diabetes (T2D) diagnosis, judge if the analysis is correct or incorrect.

You must respond with either:

- **correct**
- **incorrect**

No explanation is needed. Only output one of the two labels above.

User Prompt

Case History:

Patient is {age} y.o. {race} {ethnicity} {gender}.

Note Date: {date1}
{note-type1} note
{note-text1}

Note Date: {date2}
{note-type2} note
{note-text2}

Analysis
{Reasoner Output}

Fine-tuned LLM verifier prompt to **validate the prediction of a large reasoner** from patient demographics and sequence of clinical notes **and reasoner explanation**

Dataset Characteristics and Insights

UI Health

3332 (712 in test set) patients

Strengths: Rich longitudinal histories, diverse note types, ideal for chronic disease modeling

Limitation: Private data

MIMIC-IV

5802 (291 in test set) patients

Strength: Public data

Limitations:

- ICU-focus (omit large parts of a patient's overall hospitalization timeline)
- Narrower temporal windows limit exposure to longitudinal patterns (84.5% single visits)
- Missing temporal anchors (dates redacted for de-identification)

Key Insight: MIMIC-IV results would represent a **lower bound** for achievable real-world performance

Model Evaluations

[Details »](#)

UI-Health

- **HiT-GNN** achieves best AUC (72.2%), +4% over CLSTM (Part1)
- Best LLMs:
 - Llama-1b-ft (AUC: 65.1%)
 - ReVeAL (F1: 65.2%)

MIMIC-IV

- **HiT-GNN leads in T2D recall** (63.3%)
- Best LLM: Llama-1b-ft (AUC 64.4%, 1.5% > HiTGNN)

Note: Llama-1b is the weakest model without fine-tuning; but fine-tuning sacrifices explainability. ReVeAL provides a balanced trade-off

Conclusion & Key Contributions

Unlocking the temporal structure from clinical narratives leads to improved patient representations

Algorithmic Innovations

 **State-of-the-Art TReX methods (SpanTReX, GraphTReX):** with innovative landmark nodes

 **Hierarchical Temporal Modeling (HiT-GNN):** first application of TReX for risk detection

 **Inference-Time LLM Adaptation (ReVeAL):** scalable and robust **explainable reasoning**

Translational Value in Real Clinical Settings

 New **densely annotated sample** of real-clinical notes validating model portability

 Improving **T2D opportunistic screening** from notes, especially for **immediate risk horizon**

 **Scalable, privacy-aware framework** with rigorous data curation ensuring temporal integrity and balanced demographic representation for **transparent evaluation**.

Publications

Sequential Representation of Sparse Heterogeneous Data for Diabetes Risk Prediction

Chaturvedi R, Rashid M, Layden BT, Boyd A, Cinar A, Di Eugenio B.

IEEE BIBM 2023 (Foundational work establishing motivation for fine-grained temporal modeling)

Temporal Relation Extraction in Clinical Texts: A Span-based Graph Transformer Approach.

Chaturvedi R, Baghersshahi P, Medya S, Di Eugenio B.

ACL 2025 (SpanTrex and GraphTrex frameworks)

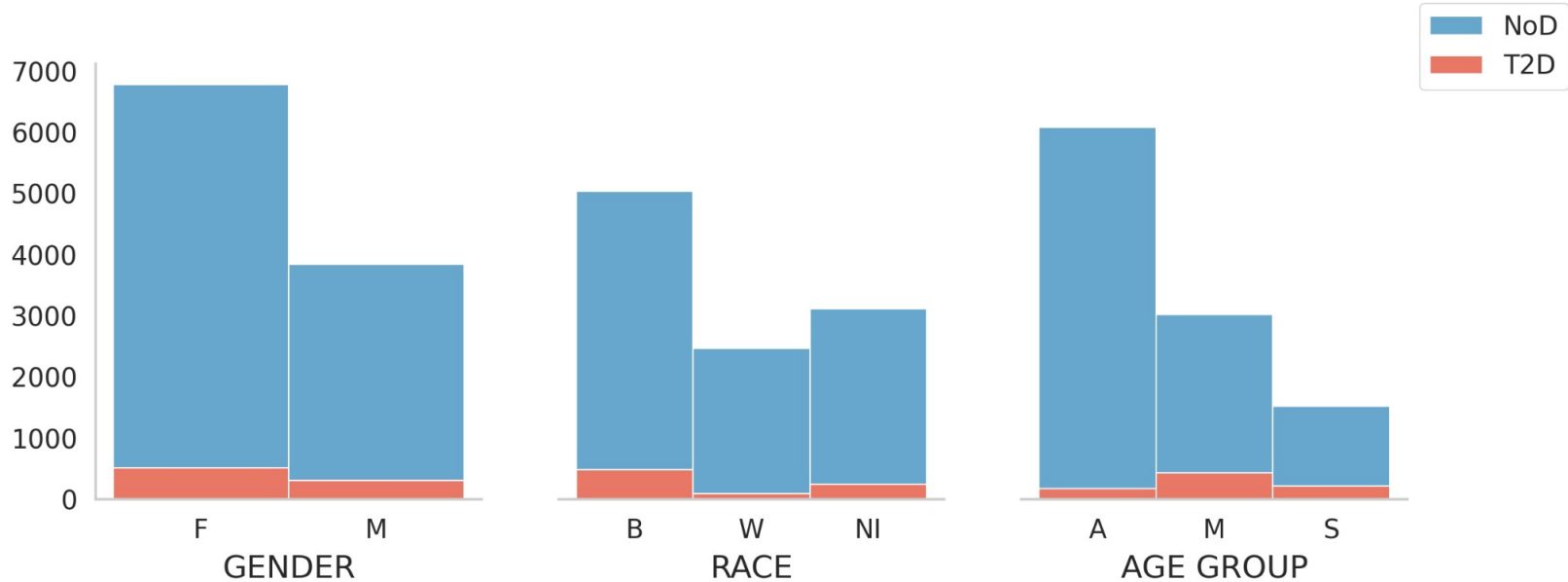
Unlocking Temporal and Semantic knowledge from Clinical Narratives for Chronic Disease prediction

Chaturvedi R, Zhou Y, Di Eugenio B.

Accepted for TACL (Early version in WWW 2024 Companion Proceedings)

Additional Materials – Parts 1 and 2

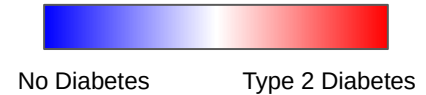
Structured Data: Demographics



- Females, and Black Americans are the majority groups across outcome classes
- Adult group are majority for NoD and Middle-age group are the majority for T2D

Interpretability Analysis

Unemployment aspirin Sedatives Gait normal Vitamins Oral Tablet Interventional procedure
Cardiovascular system Chronic kidney disease stage 5 Release (procedure) Infantile
Neuroaxonal Dystrophy Family history (finding) Anesthesia procedures Complication of



- Layerwise Relevance Propagation (LRP)^[1] Heatmap
 - Chronic kidney disease contributes highly towards positive prediction
 - Unemployment also contributes positively—an important social determinant of health

[1] S. Bach, A. Binder, G. Montavon, F. Klauschen, K.-R. Muller, and W. Samek, “On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation,” PloS one. 2015.

Ablation Across Different Encoders

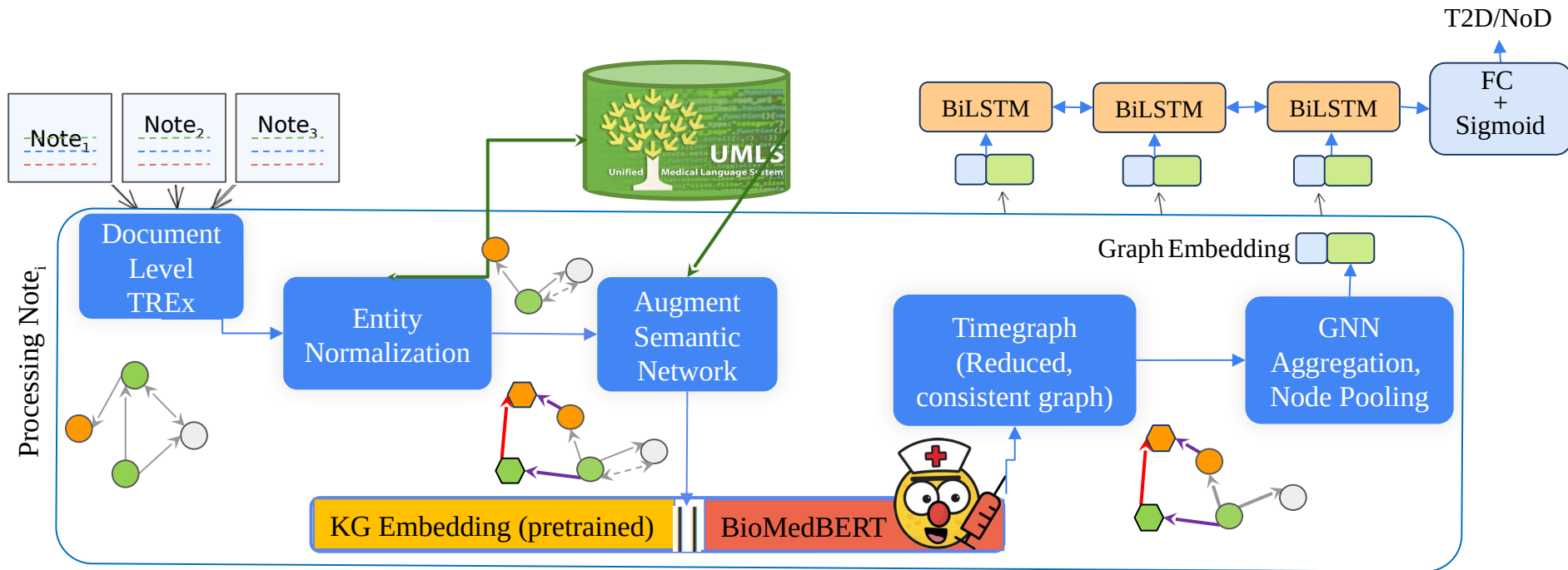
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Model	TLink (end-to-end)		
	P	R	F ₁
GRAPHTREX	78.21	61.42	68.81
BERT-SPANTREX	75.94	48.63	59.30
BERT-GRAPHTREX	77.45	52.30	62.44
RoBERTA-SPANTREX	79.98	52.05	63.06
RoBERTa-GRAPHTREX	78.22	54.88	64.50
Clinical-Longformer	82.39	52.10	63.84

Performance with Different Base Encoders on I2B2 2012

Additional Materials: Part 3

HiT-GNN



MIMIC-IV Data

- Large, **publicly available** database of **de-identified** electronic health records (EHRs)
- Patients admitted to the Beth Israel Deaconess Medical Center in Boston between 2008 and 2019.
- Successor to earlier releases MIMIC-II and MIMIC-III
- Exclusion Criteria for data curation:
 - Diagnosis of Type 1 diabetes or other types of diabetes

Data Filtering: MIMIC-IV

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* First Filtration Criteria: Only consider patients with some notes

- ▶ **Initial Data**
(excluding other types of Diabetes, RACE=Others)
- ▶ **Drop Pre-Diagnosed**
(Earliest_Dx_Date)
- ▶ **Drop if >20 Admissions** (already high risk patients)
- ▶ **Drop duplicated and non-verbal notes**
- ▶ **note length within .05-.95 quantile**
- ▶ **Drop duplicated sections**
(adjusts Earliest_dx_Date)
- ▶ **LLM + Matched Control sampling**
(adjusts Earliest_dx_Date)

Notes | T2D Patients | NoD Patients

4,033,399 | 14,101 | 142,412

3,220,793 | 5,952 | 142,412

235,727 | 5,847 | 59,566

184,421 | 5,272 | 54,803

175,201 | 5,207 | 54,170

174,995 | 5,207 | 106,222

14,957 | 2,909 | 2,909

Part-3 Results

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Model	UI-Health Corpus						MIMIC-IV					
	AUC	F ₁	T2D		NoD		AUC	F ₁	T2D		NoD	
			P	R	P	R			P	R	P	R
Llama3.2-1b	46.80	35.46	39.29	3.09	49.56	95.22	40.00	33.10	0	0	49.48	1.00
LLama3.1-8b	57.73	56.98	57.49	53.93	56.61	60.11	53.71	51.03	61.54	27.21	52.65	82.64
LLM-8b SC	55.62	55.49	56.29	50.28	55.08	60.96	52.12	43.43	66.67	12.24	51.14	93.75
Llama3.2-1b-ft	65.06	61.62	59.53	78.93	68.75	46.35	64.41	59.17	63.89	46.94	57.38	72.92
REVEAL	57.98	65.24	66.77	60.96	64.08	69.66	53.13	51.60	55.21	36.05	51.79	70.14
CLSTM	68.24	65.62	63.19	76.69	70.36	55.34	63.08	59.62	63.96	48.30	57.78	72.22
HiT-GNN	72.24	67.28	71.48	58.43	64.85	76.69	62.97	62.88	63.27	63.27	62.50	62.50
#patients	712		356		356		291		147		144	

Prompts

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System Prompt: Reasoner

You are a helpful medical assistant. Based on the patient's information and the history of medical notes from one or more visits, assess if this patient has a likelihood of being diagnosed with Type 2 diabetes in the near future.

Make the assessment based on **concrete** medical evidence and how the patient's condition has progressed. Do not solely rely on age as a risk factor.

Prediction Format:

Risk of Type 2 Diabetes: ****[True/False]**

Explanation: [Provide a brief explanation of the reasoning for the prediction.]

User Prompt

Case History:

Patient is {age} y.o. {race} {ethnicity} {gender}.

Note Date: {date1}
{note-type1} note
{note-text1}

Note Date: {date2}
{note-type2} note
{note-text2}

Note: {ethnicity} and {note-type} are excluded for the MIMIC-IV. {ethnicity} is also excluded for UI-Health corpus if value is 'NI'.

Prompt instructing an LLM to **predict future risk of Type 2 Diabetes with explanation** from patient demographics and sequence of clinical notes

System Prompt: Fine-tuned Reasoner

You are a helpful medical assistant. Based on the patient's information and the history of medical notes from one or more visits, assess if this patient has likelihood of being diagnosed with Type 2 diabetes in the near future.

You must respond with either:

- `true`
- `false`

No explanation is needed. Only output one of the two labels above.

Fine-tuned LLM prompt to **predict future risk of Type 2 Diabetes**
from patient demographics and sequence of clinical notes